Detecting Hidden Data Leakage Without Knowing the Pipeline: An Automated Framework Based on Dataset and Model Behavior

Jackfruit Project — One-Pager

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Project Summary

ML results are often inflated by data leakage, yet finding it normally requires full knowledge of the training pipeline. This project builds an automated detector that works from the data and the trained model alone, and measures how reliably it finds four controlled leakage types in multiple datasets and models.

- **Leakage types (4):** (1) target leakage: a feature encodes or is derived from the label; (2) train–test contamination: test rows duplicated or near-duplicated in training; (3) temporal leakage: training uses data from after the test period; (4) preprocessing leakage: scaling, imputation or encoding fitted on train + test together.
- **Detector input:** the processed train and test splits exactly as given to the model (features, labels, optional timestamp column) and the trained model as a black box (predict / predict_proba only). No code, logs or pipeline details.
- **Detector output:** for each of the four types, a leakage score (0–1) and a binary flag, plus an overall verdict (clean / leaky), the evidence behind each flag (suspect features, duplicate row IDs, shifted time boundary) and an estimated performance inflation.
- **Signals and evaluation:** feature–target mutual information, duplicate overlap, adversarial validation, time-ordered re-splits, feature ablation and train–test confidence gaps. Scored on detection accuracy, false-positive rate, false-negative rate and performance inflation (leaky vs. clean score) across Logistic Regression, Random Forest, XGBoost and an MLP.

Possible Datasets

- Adult Income (UCI) and Telco Customer Churn — tabular classification, suited to target-leakage injection.
- Credit Card Fraud (ULB/Kaggle) — imbalanced data, sensitive to duplicates and contamination.
- Household Power Consumption / Rossmann Store Sales — time-ordered data for temporal leakage.
- California Housing and Forest Covertype — regression and multiclass settings to test generalization of the detector.

Research Papers

1. Kapoor, S., & Narayanan, A. (2023). “Leakage and the reproducibility crisis in machine-learning-based science.” Patterns, 4(9), 100804. DOI: 10.1016/j.patter.2023.100804. **Relevance:** Provides a taxonomy of eight leakage types, which guides the scenarios we inject.

2. Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). “Leakage in data mining: Formulation, detection, and avoidance.” ACM TKDD, 6(4), Article 15. DOI: 10.1145/2382577.2382579. **Relevance:** Formalizes leakage and discusses detection when the modeler has no control over data collection.

Novelty

Existing tools and guidelines mostly assume access to the code or preprocessing steps. This project targets pipeline-agnostic detection, combining data-level, statistical, and model-level signals in one framework that covers four precisely defined leakage types. Controlled leakage injection allows a systematic benchmark of false-positive and false-negative rates and a direct measure of the performance inflation each leak causes.

Expected Learnings

- How each leakage type inflates performance and which signals expose it most reliably.
- Limits of pipeline-free detection, i.e., which leaks (e.g., subtle preprocessing leakage) stay hidden.
- Designing controlled experiments and measuring detectors with FPR, FNR, and accuracy.
- Leakage-free evaluation practices: proper splitting, time-aware validation, and deduplication.